Khashayar Pourtaheri

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Education

University of Washington Expected 2027
Ph.D. in Economics | Advisor: Alan Griffith

Smith School of Business, Queen's University 2021
M.Sc. in Management, Business Economics

Sharif University of Technology 2020
M.Sc. in Economics

Sharif University of Technology 2017
B.Sc. in Mechanical Engineering

Research

Fields: Applied Microeconomics, Health and Labor Economics, Causal Inference, Behavioral Economics and AI

Job Market Paper

The Social Cost of Drug Decriminalization [link]
SSRN working paper October 2026

- Estimates the social costs and benefits of decriminalizing drug possession in Oregon (Measure 110) and Washington (Blake), and of later restoring penalties.
- Applies synthetic difference-in-differences to FBI arrest and crime records from nearly 5,000 agencies and CDC overdose records, addressing the literature's debate over fentanyl and COVID by matching comparison groups on fentanyl's arrival, the pandemic, and politics.
- Finds that drug-user arrests fall 68%, while overdose deaths rise 34% (1,971 excess deaths) and property crime 17%, for a net social cost of $26.6 billion by mid-2023; restoring penalties in Washington brings arrests back but not deaths down.
- Estimates heterogeneous effects with synthetic DiD and cross-fitted DML-DiD: property crime rises more west of the Cascades than east (19% vs. 8%), violent offenses rise more among suspected drug users than other offenders (23% vs. 5%), and the Black–White arrest gap closes in levels but not in ratio.

Working Papers

Behavioral History Outperforms Descriptions of the Person for LLM Synthetic Personas [link]
with Ahmad Zareei | arXiv preprint October 2026

- Builds 4,225 synthetic personas with Claude, one for each of 845 U.S. adults under five nested information conditions, to predict each person's choices on 14 behavioral biases.
- Finds that behavioral history beats every description of the person: demographics, personality, and cognitive scores, even combined, reach 7–12% of the human test–retest benchmark, while adding past answers raises this to 28%, the best in all 17 demographic groups.
- Shows that population averages look right but mask individual-level failure, and that personas overweight education and income (74% vs. 18% of explained variation).

Work in Progress

- **The Effects of Common Ownership on the Oligopoly**
  Contract-theoretic model of the conditions under which common-ownership links can sustain collusive product-market outcomes.
- **The Price Effect of the "Salon" Mechanism on the Price of Cargo**
  Reduced-form analysis of 40 million bills of lading estimating the price effects of government intervention and information disclosure in the freight market.

Research Experience

Research Economist — Adolescent Mental Health Study, CHOICE Institute Sep 2026 – Present
Supervisor: Noémi Kreif

- Evaluate the causal design and outcome measurement for a study of how adolescent depression affects adult hours, participation, earnings, and education, using restricted-use Add Health data and survey-weighted, cross-fitted TMLE.
- Audited the R estimation pipeline and outcome coding, documenting identification threats and coding errors and proposing diagnostics and sensitivity analyses (E-values, MNAR).
- Translated treatment effects into labor-supply elasticities for downstream ROI models, and re-estimated 17 NLSY97 outcomes with cross-fitted AIPW as a public-data benchmark.

Data Scientist Researcher — Institute for Health Metrics and Evaluation (IHME) Summer 2025

- Produced subnational estimates of HIV prevalence, antiretroviral-therapy coverage, and viral-load suppression for Malawi and Zimbabwe by integrating PHIA, DHS, and AIS survey data in reproducible pipelines.
- Conducted statistical, machine-learning, and forecasting analyses; documented legacy code and mapped the end-to-end workflow.

Research Assistant — Smith School of Business, Queen's University May 2021 – Aug 2021

- Integrated and cleaned data from multiple sources, validated data quality, and communicated statistical findings through reproducible analysis in R and Stata.

Teaching

Instructor — Intro to Microeconomics, University of Washington Winter 2024

- Designed and delivered lectures on supply and demand, market equilibrium, consumer theory, and firm behavior; created assessments and held office hours.

Teaching Assistant — University of Washington Sep 2022 – Present

- Courses: econometrics and data science; probability and statistics; introductory micro- and macroeconomics.
- Taught R and Python labs on data cleaning, visualization, reproducible workflows, supervised learning, cross-validation, regularization, tree-based models, and neural networks.

Teaching Assistant — Managerial Economics (MBA), Queen's University Jan 2021 – May 2021

- Led quiz and review sessions for MBA students, translating economic concepts for working professionals.

Fellowships & Honors

Jeff McGill Graduate Fellowship, Queen's University 2021
Ranked 8th, National Scientific Olympiad in Economics 2017
Top 0.1% of approximately 300,000 national university-entrance applicants 2012

Skills

Econometrics: Event-study DiD, synthetic DiD and synthetic control, DML, CV-TMLE, AIPW, and Super Learner, IV, RDD, survey-weighted inference, mediation and sensitivity analysis (E-values, MNAR)
Programming: R (tmle, SuperLearner, survey), Stata, Python, SQL, Git, MATLAB, LaTeX/Beamer
Data: Add Health (restricted-use), NLSY97, FBI NIBRS, CDC WONDER, PHIA, DHS, AIS
Other: LLM experiment design (Claude API), machine learning (scikit-learn, PyTorch), Tableau